Sample Project Documentation

Phase 7: Project Documentation

Abstract

trvale-book is a web-based travel planning application that creates a structured, budget-aware itinerary from a traveller's preferences. A five-step wizard collects destination, dates, travellers, budget, style, pace, interests and dietary needs. A FastAPI backend validates the request, obtains a plan from Google Gemini or OpenAI (or a built-in fallback generator) and computes a budget breakdown. The React frontend lets the user review, edit, save and print the plan, and Supabase provides authentication and row-level-secured storage.

1. Introduction

Planning a trip means balancing time, cost and interests. A list of attractions is not enough: the traveller also needs a day-wise schedule that fits the budget and can be changed. trvale-book addresses the first planning step by producing an organized draft from a single guided form.

2. Objectives

- Reduce the effort needed to create a first itinerary draft.
- Produce morning, afternoon and evening activities with estimated costs.
- Show a live budget comparison and warnings.
- Allow editing, regeneration of a day, saving, searching and deleting.
- Provide secure sign-in and per-user data access.
- Support printing and saving the plan as PDF.

3. Technology and Architecture

The frontend uses React 19, Vite and Tailwind CSS with React Router. The backend uses Python 3.11 and FastAPI with Pydantic schemas. The AI layer supports Gemini and OpenAI. Supabase provides authentication and PostgreSQL with row-level security. The frontend is deployed on Vercel and the backend is prepared for Render.

4. Methodology

1. Collect preferences through the wizard.
2. Send them to the generate endpoint.
3. Validate the request with Pydantic.
4. Build a prompt and call the AI provider, or use the fallback generator.
5. Hydrate the reply and compute the budget breakdown.
6. Return the trip to the frontend and display it.
7. Let the user edit, save, filter, delete and print.

5. Functional Modules

The authentication module handles Supabase and demo sessions; the wizard module collects inputs; the AI module generates itineraries; the budget module calculates costs and warnings; the trip module lists, edits and deletes trips; the profile and settings modules store defaults and diagnostics; and the print module formats the output.

6. Testing and Evaluation

The repository contains 17 backend and 6 frontend automated tests and a manual end-to-end checklist. Results must be recorded from actual runs. The documentation does not claim a completed benchmark or user study.

7. Limitations

AI-generated recommendations and costs are estimates. Booking, payment and live provider data are not implemented. The current backend keeps generated and demo trips in memory and writes only the trip row to Supabase on creation, so full persistence of days and activities needs further work. Day regeneration is a lightweight transformation and not a new model call.

8. Future Scope

Planned improvements include complete database persistence, model-backed day regeneration, automated end-to-end tests, approved live place and weather data, authorized booking integrations, multilingual support, accessibility improvements, rate limiting and monitoring.

9. Conclusion

trvale-book demonstrates an end-to-end AI-assisted planning workflow with budget analysis, editing and printing. Additional testing and evidence capture are needed before making production-level reliability claims.